KUTAISI INTERNATIONAL UNIVERSITY

Scientific Computing

Ramaz Botchorishvili

Kutaisi International University

#1.2

Mathematical models with ODEs

- Population dynamics
- Radiation decay
- Chemical kietic
- SIR models
- Predator-prey
- Investment consumption model
- Ball motion
- Many body problem
- Helmhotz equation
- Schroedinger equation
- Swinging fabric
- Q & A

Population growth

Problem 1.1

- $N(t)$ number of individuals at time t

Population growth

Problem 1.1

- $N(t)$ number of individuals at time t
- Predict evolution of $N(t)$ in time

Population growth

Problem 1.1

- $N(t)$ number of individuals at time t
- Predict evolution of $N(t)$ in time

Derivation of the model

- $N(t + \Delta t)$ - number of individuals at time $t + \Delta t$

Population growth

Problem 1.1

- $N(t)$ number of individuals at time t
- Predict evolution of $N(t)$ in time

Derivation of the model

- $N(t + \Delta t)$ - number of individuals at time $t + \Delta t$
- $N(t)$ number of individuals at time t

Population growth

Problem 1.1

- $N(t)$ number of individuals at time t
- Predict evolution of $N(t)$ in time

Derivation of the model

- $N(t+\Delta t)$ - number of individuals at time $t+\Delta t$
- $N(t)$ number of individuals at time t
- $bN(t)$ - number of newborns, b - number

Population growth

Problem 1.1

- $N(t)$ number of individuals at time t
- Predict evolution of $N(t)$ in time

Derivation of the model

- $N(t+\Delta t)$ - number of individuals at time $t+\Delta t$
- $N(t)$ number of individuals at time t
- $bN(t)$ - number of newborns, b - number
- $dN(t)$ - number of deaths, d - number

Population growth

Problem 1.1

- $N(t)$ number of individuals at time t
- Predict evolution of $N(t)$ in time

Derivation of the model

- $N(t+\Delta t)$ - number of individuals at time $t+\Delta t$
- $N(t)$ number of individuals at time t
- $bN(t)$ - number of newborns, b - number
- $dN(t)$ - number of deaths, d - number
- $N(t+\Delta t)-N(t)$ - net growth of the population

Population growth

Problem 1.1

- $N(t)$ number of individuals at time t
- Predict evolution of $N(t)$ in time

Derivation of the model

- $N(t+\Delta t)$ - number of individuals at time $t+\Delta t$
- $N(t)$ number of individuals at time t
- $bN(t)$ - number of newborns, b - number
- $dN(t)$ - number of deaths, d - number
- $N(t+\Delta t)-N(t)$ - net growth of the population
- $\int_t^{t+\Delta t}[bN(\tau)-dN(\tau)]d\tau$ - net growth of the population

Population growth

Problem 1.1

- $N(t)$ number of individuals at time t
- Predict evolution of $N(t)$ in time

Derivation of the model

- $N(t+\Delta t)$ - number of individuals at time $t+\Delta t$
- $N(t)$ number of individuals at time t
- $bN(t)$ - number of newborns, b - number
- $dN(t)$ - number of deaths, d - number
- $N(t+\Delta t)-N(t)$ - net growth of the population
- $\int_t^{t+\Delta t}[bN(\tau)-dN(\tau)]d\tau$ - net growth of the population
- $N(t+\Delta t)-N(t)=\int_t^{t+\Delta t}[bN(\tau)-dN(\tau)]d\tau$
- $\frac{1}{\Delta t}(N(t+\Delta t)-N(t))=\frac{1}{\Delta t}\int_t^{t+\Delta t}[bN(\tau)-dN(\tau)]d\tau$

Population growth

Problem 1.1

- $N(t)$ number of individuals at time t
- Predict evolution of $N(t)$ in time

Derivation of the model

- $N(t+\Delta t)$ - number of individuals at time $t+\Delta t$
- $N(t)$ number of individuals at time t
- $bN(t)$ - number of newborns, b - number
- $dN(t)$ - number of deaths, d - number
- $N(t+\Delta t)-N(t)$ - net growth of the population
- $\int_t^{t+\Delta t}[bN(\tau)-dN(\tau)]d\tau$ - net growth of the population
- $N(t+\Delta t)-N(t)=\int_t^{t+\Delta t}[bN(\tau)-dN(\tau)]d\tau$
- $\frac{1}{\Delta t}(N(t+\Delta t)-N(t))=\frac{1}{\Delta t}\int_t^{t+\Delta t}[bN(\tau)-dN(\tau)]d\tau$
  $\Downarrow \ \Delta t\to 0$

Population growth

Problem 1.1

- $N(t)$ number of individuals at time t
- Predict evolution of $N(t)$ in time

Derivation of the model

- $N(t+\Delta t)$ - number of individuals at time $t+\Delta t$
- $N(t)$ number of individuals at time t
- $bN(t)$ - number of newborns, b - number
- $dN(t)$ - number of deaths, d - number
- $N(t+\Delta t)-N(t)$ - net growth of the population
- $\int_t^{t+\Delta t}[bN(\tau)-dN(\tau)]d\tau$ - net growth of the population
- $N(t+\Delta t)-N(t)=\int_t^{t+\Delta t}[bN(\tau)-dN(\tau)]d\tau$
- $\frac{1}{\Delta t}(N(t+\Delta t)-N(t))=\frac{1}{\Delta t}\int_t^{t+\Delta t}[bN(\tau)-dN(\tau)]d\tau$

$$\Downarrow \quad \Delta t \to 0$$

- $\frac{dN(t)}{dt}=rN(t), r=b-d$

Population growth

Differential equation for modeling population growth

- $\frac{dN(t)}{dt} = rN(t), r = b - d$

Population growth

Differential equation for modeling population growth

- $\frac{dN(t)}{dt} = rN(t), r = b - d$
- Unknown - function $N(t)$ number of individuals at time t

Population growth

Differential equation for modeling population growth

- $\frac{dN(t)}{dt} = rN(t), r = b - d$
- Unknown - function $N(t)$ number of individuals at time t

Predict evolution of $N(t)$ in time

- $N(t) = Ce^{rt}$, C - arbitrary constant

Population growth

Differential equation for modeling population growth

- $\frac{dN(t)}{dt} = rN(t), r = b - d$
- Unknown - function $N(t)$ number of individuals at time t

Predict evolution of $N(t)$ in time

- $N(t) = Ce^{rt}$, C - arbitrary constant
- Suppose $N(0) = N_0$

Population growth

Differential equation for modeling population growth

- $\frac{dN(t)}{dt} = rN(t), r = b - d$
- Unknown - function $N(t)$ number of individuals at time t

Predict evolution of $N(t)$ in time

- $N(t) = Ce^{rt}$, C - arbitrary constant
- Suppose $N(0) = N_0$
- $N(t) = N_0 e^{rt}$ - number of individuals at time t

Population growth

Differential equation for modeling population growth

- $\frac{dN(t)}{dt} = rN(t), r = b - d$
- Unknown - function $N(t)$ number of individuals at time t

Predict evolution of $N(t)$ in time

- $N(t) = Ce^{rt}$, C - arbitrary constant
- Suppose $N(0) = N_0$
- $N(t) = N_0 e^{rt}$ - number of individuals at time t

$$\Downarrow$$

Population growth

Differential equation for modeling population growth

- $\frac{dN(t)}{dt} = rN(t), r = b - d$
- Unknown - function $N(t)$ number of individuals at time t

Predict evolution of $N(t)$ in time

- $N(t) = Ce^{rt}$, C - arbitrary constant
- Suppose $N(0) = N_0$
- $N(t) = N_0 e^{rt}$ - number of individuals at time t

$$\Downarrow$$

  - Initial condition at some time, usually at $t = 0$ is needed

Population growth

Differential equation for modeling population growth

- $\frac{dN(t)}{dt} = rN(t), r = b - d$
- Unknown - function $N(t)$ number of individuals at time t

Predict evolution of $N(t)$ in time

- $N(t) = Ce^{rt}$, C - arbitrary constant
- Suppose $N(0) = N_0$
- $N(t) = N_0 e^{rt}$ - number of individuals at time t

$$\Downarrow$$

  - Initial condition at some time, usually at $t = 0$ is needed

Problem 1.2

Solution of the problem 1.1

Population growth

Differential equation for modeling population growth

- $\frac{dN(t)}{dt} = rN(t), r = b - d$
- Unknown - function $N(t)$ number of individuals at time t

Predict evolution of $N(t)$ in time

- $N(t) = Ce^{rt}$, C - arbitrary constant
- Suppose $N(0) = N_0$
- $N(t) = N_0 e^{rt}$ - number of individuals at time t

$$\Downarrow$$

  - Initial condition at some time, usually at $t = 0$ is needed

Problem 1.2

Solution of the problem 1.1

$$\frac{dN(t)}{dt} = rN(t), N(0) = N_0$$

Radioactive decay

Problem 1.3

- $y(t)$ mass of radioactive substance at time t

Radioactive decay

Problem 1.3

- $y(t)$ mass of radioactive substance at time t
- Law of radioactive decay: rate of decay is proportional to $y(t)$

Radioactive decay

Problem 1.3

- $y(t)$ mass of radioactive substance at time t
- Law of radioactive decay: rate of decay is proportional to $y(t)$
- Predict evolution of $y(t)$ in time

Radioactive decay

Problem 1.3

- $y(t)$ mass of radioactive substance at time t
- Law of radioactive decay: rate of decay is proportional to $y(t)$
- Predict evolution of $y(t)$ in time

Equation of radioactive decay

$$\frac{dy(t)}{dt} = -ry(t), r > 0, t > 0$$

$$y(0) = y_0$$

Radioactive decay

Problem 1.3

- $y(t)$ mass of radioactive substance at time t
- Law of radioactive decay: rate of decay is proportional to $y(t)$
- Predict evolution of $y(t)$ in time

Equation of radioactive decay

$$\frac{dy(t)}{dt} = -ry(t), r > 0, t > 0$$

$$y(0) = y_0$$

$$y(t) = y_0 e^{-rt}$$

Chemical kinetic

Problem 1.4

Reversible reaction of two substances

- $A(t), B(t)$ - concentration of two substances at t

Chemical kinetic

Problem 1.4

Reversible reaction of two substances

- $A(t), B(t)$ - concentration of two substances at t
- $A(t)$ is turned into $B(t)$ at a rate k_A

Chemical kinetic

Problem 1.4

Reversible reaction of two substances

- $A(t), B(t)$ - concentration of two substances at t
- $A(t)$ is turned into $B(t)$ at a rate k_A
- $B(t)$ is turned into $A(t)$ at a rate k_A

Chemical kinetic

Problem 1.4

Reversible reaction of two substances

- $A(t), B(t)$ - concentration of two substances at t
- $A(t)$ is turned into $B(t)$ at a rate k_A
- $B(t)$ is turned into $A(t)$ at a rate k_A
- Predict evolution of $A(t), B(t)$ in time

Chemical kinetic

Problem 1.4

Reversible reaction of two substances

- $A(t), B(t)$ - concentration of two substances at t
- $A(t)$ is turned into $B(t)$ at a rate k_A
- $B(t)$ is turned into $A(t)$ at a rate k_A
- Predict evolution of $A(t), B(t)$ in time

Initial value problem, H.P. Langtagen, Springer 2016

$$\frac{dA(t)}{dt} = -k_A A(t) + k_B B(t)$$

$$\frac{dB(t)}{dt} = k_A A(t) - k_B B(t)$$

$$A(0) = A_0, B(0) = B_0$$

Chemical kinetic

Problem 1.4

Reversible reaction of two substances

- $A(t), B(t)$ - concentration of two substances at t
- $A(t)$ is turned into $B(t)$ at a rate k_A
- $B(t)$ is turned into $A(t)$ at a rate k_A
- Predict evolution of $A(t), B(t)$ in time

Initial value problem, H.P. Langtagen, Springer 2016

$$\frac{dA(t)}{dt} = -k_A A(t) + k_B B(t)$$

$$\frac{dB(t)}{dt} = k_A A(t) - k_B B(t)$$

$$A(0) = A_0, B(0) = B_0$$

$$\frac{dA(t)}{dt} + \frac{dB(t)}{dt} = 0$$

Chemical kinetic

Problem 1.4

Reversible reaction of two substances

- $A(t), B(t)$ - concentration of two substances at t
- $A(t)$ is turned into $B(t)$ at a rate k_A
- $B(t)$ is turned into $A(t)$ at a rate k_A
- Predict evolution of $A(t), B(t)$ in time

Initial value problem, H.P. Langtagen, Springer 2016

$$\frac{dA(t)}{dt} = -k_A A(t) + k_B B(t)$$

$$\frac{dB(t)}{dt} = k_A A(t) - k_B B(t)$$

$$A(0) = A_0, B(0) = B_0$$

$$\frac{dA(t)}{dt} + \frac{dB(t)}{dt} = 0 \Rightarrow A(t) + B(t) = A_0 + B_0$$

Chemical kinetic

Initial value problem

Irreversible reaction of two substances into third, , H.P. Langtagen, Springer 2016

$$\frac{dC(t)}{dt} = kA(t)B(t)$$

$$\frac{dA(t)}{dt} = -kA(t)B(t)$$

$$\frac{dB(t)}{dt} = -kA(t)B(t)$$

$$A(0) = A_0, B(0) = B_0, C(0) = C_0$$

$$k > 0$$

Spreading diseases

Problem 1.5

- $S(t)$ number of individuals susceptible to infection at time t

Spreading diseases

Problem 1.5

- $S(t)$ number of individuals susceptible to infection at time t
- $I(t)$ number of individuals infected at time t

Spreading diseases

Problem 1.5

- $S(t)$ number of individuals susceptible to infection at time t
- $I(t)$ number of individuals infected at time t
- $R(t)$ number of individuals died at time t

Spreading diseases

Problem 1.5

- $S(t)$ number of individuals susceptible to infection at time t
- $I(t)$ number of individuals infected at time t
- $R(t)$ number of individuals died at time t
- Predict evolution of $S(t), I(t), R(t)$ in time

Spreading diseases

Problem 1.5

- $S(t)$ number of individuals susceptible to infection at time t
- $I(t)$ number of individuals infected at time t
- $R(t)$ number of individuals died at time t
- Predict evolution of $S(t), I(t), R(t)$ in time

$S(t)I(t)$ - number of possible (susceptible, infected) pairs

Initial value problem, H.P. Langtagen, Springer 2016

$$\frac{dS(t)}{dt} = -\beta S(t)I(t)$$

$$\frac{dI(t)}{dt} = \beta S(t)I(t) - \gamma I(t)$$

$$\frac{dR(t)}{dt} = \gamma I(t)$$

$$S(0) = S_0, I(0) = I_0, R(0) = 0, \beta > 0, \gamma > 0$$

Spreading diseases

Problem 1.5

- $S(t)$ number of individuals susceptible to infection at time t
- $I(t)$ number of individuals infected at time t
- $R(t)$ number of individuals died at time t
- Predict evolution of $S(t), I(t), R(t)$ in time

$S(t)I(t)$ - number of possible (susceptible, infected) pairs

Initial value problem, H.P. Langtagen, Springer 2016

$$\frac{dS(t)}{dt} = -\beta S(t)I(t)$$

$$\frac{dI(t)}{dt} = \beta S(t)I(t) - \gamma I(t)$$

$$\frac{dR(t)}{dt} = \gamma I(t)$$

$$S(0) = S_0, I(0) = I_0, R(0) = 0, \beta > 0, \gamma > 0$$

$$S(t) + I(t) + R(t) = const$$

Spreading diseases

SIR with time-restricted immunity, , H.P. Langtagen, Springer 2016

$$SIR : S \to I \to R$$

SIR with time restricted immunity: $S \to I \to R \to S$

$$\frac{dS(t)}{dt} = -\beta S(t)I(t) + \nu R(t)$$

$$\frac{dI(t)}{dt} = \beta S(t)I(t) - \gamma I(t)$$

$$\frac{dR(t)}{dt} = \gamma I(t) - \nu R(t)$$

$$S(0) = S_0, I(0) = I_0, R(0) = 0,$$

$$\beta > 0, \gamma > 0, \nu > 0$$

Spreading diseases

SIR with time-restricted immunity, , H.P. Langtagen, Springer 2016

$$SIR : S \to I \to R$$

SIR with time restricted immunity: $S \to I \to R \to S$

$$\frac{dS(t)}{dt} = -\beta S(t)I(t) + \nu R(t)$$

$$\frac{dI(t)}{dt} = \beta S(t)I(t) - \gamma I(t)$$

$$\frac{dR(t)}{dt} = \gamma I(t) - \nu R(t)$$

$$S(0) = S_0, I(0) = I_0, R(0) = 0,$$

$$\beta > 0, \gamma > 0, \nu > 0$$

$$S(t) + I(t) + R(t) = const$$

Spreading diseases

SIRV - incorporating vaccination, , H.P. Langtagen, Springer 2016

$$\frac{dS(t)}{dt} = -\beta S(t)I(t) + \nu R(t) - pS(t)$$

$$\frac{dV(t)}{dt} = pS(t)$$

$$\frac{dI(t)}{dt} = \beta S(t)I(t) - \gamma I(t)$$

$$\frac{dR(t)}{dt} = \gamma I(t) - \nu R(t)$$

$$S(0) = S_0, I(0) = I_0, R(0) = 0,$$

$$\beta > 0, \gamma > 0, \nu > 0, p > 0$$

$$S(t) + V(t) + I(t) + R(t) = const$$

Predator-Prey model in ecology

Problem 1.6

- $L(t)$ number of predators at time t, e.g. lynx

Predator-Prey model in ecology

Problem 1.6

- $L(t)$ number of predators at time t, e.g. lynx
- $H(t)$ number of preys at time t, e.g hares

Predator-Prey model in ecology

Problem 1.6

- $L(t)$ number of predators at time t, e.g. lynx
- $H(t)$ number of preys at time t, e.g hares
- Predict evolution of $L(t), H(t)$ in time

Predator-Prey model in ecology

Problem 1.6

- $L(t)$ number of predators at time t, e.g. lynx
- $H(t)$ number of preys at time t, e.g hares
- Predict evolution of $L(t), H(t)$ in time

$L(t)H(t)$ - number of possible (predator, prey) pairs

Predator-Prey model in ecology

Problem 1.6

- $L(t)$ number of predators at time t, e.g. lynx
- $H(t)$ number of preys at time t, e.g hares
- Predict evolution of $L(t), H(t)$ in time

$L(t)H(t)$ - number of possible (predator, prey) pairs

Initial value problem, H.P. Langtagen, Springer 2016

$$\frac{dH(t)}{dt} = aH(t) - bH(t)L(t)$$

$$\frac{dL(t)}{dt} = dH(t)L(t) - cL(t)$$

$$H(0) = H_0, L(0) = L_0,$$

$$a, b, c, d > 0$$

Predator-Prey model in ecology

Problem 1.6

- $L(t)$ number of predators at time t, e.g. lynx
- $H(t)$ number of preys at time t, e.g hares
- Predict evolution of $L(t), H(t)$ in time

$L(t)H(t)$ - number of possible (predator, prey) pairs

Initial value problem, H.P. Langtagen, Springer 2016

$$\frac{dH(t)}{dt} = aH(t) - bH(t)L(t)$$

$$\frac{dL(t)}{dt} = dH(t)L(t) - cL(t)$$

$$H(0) = H_0, L(0) = L_0,$$

$$a, b, c, d > 0$$

- a - population of hares increases

Predator-Prey model in ecology

Problem 1.6

- $L(t)$ number of predators at time t, e.g. lynx
- $H(t)$ number of preys at time t, e.g hares
- Predict evolution of $L(t), H(t)$ in time

$L(t)H(t)$ - number of possible (predator, prey) pairs

Initial value problem, H.P. Langtagen, Springer 2016

$$\frac{dH(t)}{dt} = aH(t) - bH(t)L(t)$$

$$\frac{dL(t)}{dt} = dH(t)L(t) - cL(t)$$

$$H(0) = H_0, L(0) = L_0,$$

$$a, b, c, d > 0$$

- a - population of hares increases
- d - population of lynx increases when they eat prey

Predator-Prey model in ecology

Problem 1.6

- $L(t)$ number of predators at time t, e.g. lynx
- $H(t)$ number of preys at time t, e.g hares
- Predict evolution of $L(t), H(t)$ in time

$L(t)H(t)$ - number of possible (predator, prey) pairs

Initial value problem, H.P. Langtagen, Springer 2016

$$\frac{dH(t)}{dt} = aH(t) - bH(t)L(t)$$

$$\frac{dL(t)}{dt} = dH(t)L(t) - cL(t)$$

$$H(0) = H_0, L(0) = L_0,$$

$$a, b, c, d > 0$$

- a - population of hares increases
- d - population of lynx increases when they eat prey
- b - population of hares decreases when they meet predator

Predator-Prey model in ecology

Problem 1.6

- $L(t)$ number of predators at time t, e.g. lynx
- $H(t)$ number of preys at time t, e.g hares
- Predict evolution of $L(t), H(t)$ in time

$L(t)H(t)$ - number of possible (predator, prey) pairs

Initial value problem, H.P. Langtagen, Springer 2016

$$\frac{dH(t)}{dt} = aH(t) - bH(t)L(t)$$

$$\frac{dL(t)}{dt} = dH(t)L(t) - cL(t)$$

$$H(0) = H_0, L(0) = L_0,$$

$$a, b, c, d > 0$$

- a - population of hares increases
- d - population of lynx increases when they eat prey
- b - population of hares decreases when they meet predator
- c - population of lynx decreases

Market Competition

Cmpetition between two companies in a duopoly, where one company's growth can come at the expense of the other.

$$\frac{dx}{dt} = ax - bxy$$
$$\frac{dy}{dt} = -cy + dxy$$
$$x(0) = x_0, \quad y(0) = y_0$$

Market Competition

Cmpetition between two companies in a duopoly, where one company's growth can come at the expense of the other.

$$\frac{dx}{dt} = ax - bxy$$
$$\frac{dy}{dt} = -cy + dxy$$
$$x(0) = x_0, \quad y(0) = y_0$$

- $x(t)$: Population (or market share) of company X (the "prey")
- $y(t)$: Population (or market share) of company Y (the "predator")
- a, b, c, d: Positive constants representing growth rates and interaction rates.

Market Competition

Cmpetition between two companies in a duopoly, where one company's growth can come at the expense of the other.

$$\frac{dx}{dt} = ax - bxy$$
$$\frac{dy}{dt} = -cy + dxy$$
$$x(0) = x_0, \quad y(0) = y_0$$

- $x(t)$: Population (or market share) of company X (the "prey")
- $y(t)$: Population (or market share) of company Y (the "predator")
- a, b, c, d: Positive constants representing growth rates and interaction rates.
- Growth term: ax
- Decline term: $-cy$
- Interaction term, negative impact of Y on X: $-bxy$
- Interaction term, positive impact of X on Y: dxy

Financial Planning

Different scenarios which accounts for income, interest rates, consumption and adjustment measures

Financial Planning

Different scenarios which accounts for income, interest rates, consumption and adjustment measures

$$\frac{dA}{dt} = rA - C$$
$$\frac{dC}{dt} = \alpha(A - A_d)$$
$$x(0) = x_0, \quad y(0) = y_0$$

Financial Planning

Different scenarios which accounts for income, interest rates, consumption and adjustment measures

$$\frac{dA}{dt} = rA - C$$
$$\frac{dC}{dt} = \alpha(A - A_d)$$
$$x(0) = x_0, \quad y(0) = y_0$$

- Investment amount: $A(t)$
- Consumption: $C(t)$
- Interest rate: r
- Adjustment paramete: α
- Desired investment level: A_d

Financial Planning

Different scenarios which accounts for income, interest rates, consumption and adjustment measures

$$\frac{dA}{dt} = rA - C$$
$$\frac{dC}{dt} = \alpha(A - A_d)$$
$$x(0) = x_0, \quad y(0) = y_0$$

- Investment amount: $A(t)$
- Consumption: $C(t)$
- Interest rate: r
- Adjustment paramete: α
- Desired investment level: A_d
- Investment growth: $rA - C$
- Adjustment of conumption: $\alpha(A - A_d)$

Ball Motion

$$\frac{dx}{dt} = v_x$$
$$\frac{dy}{dt} = v_y$$
$$\frac{dv_x}{dt} = -\frac{k}{m} v_x \sqrt{v_x^2 + v_y^2}$$
$$\frac{dv_y}{dt} = -g - \frac{k}{m} v_y \sqrt{v_x^2 + v_y^2}$$

Ball Motion

$$\frac{dx}{dt} = v_x$$
$$\frac{dy}{dt} = v_y$$
$$\frac{dv_x}{dt} = -\frac{k}{m} v_x \sqrt{v_x^2 + v_y^2}$$
$$\frac{dv_y}{dt} = -g - \frac{k}{m} v_y \sqrt{v_x^2 + v_y^2}$$

$$x(0) = x_0, \quad y(0) = y_0, \quad v_x(0) = v_{x0}, \quad v_y(0) = v_{y0}$$

Ball Motion

$$\frac{dx}{dt} = v_x$$
$$\frac{dy}{dt} = v_y$$
$$\frac{dv_x}{dt} = -\frac{k}{m} v_x \sqrt{v_x^2 + v_y^2}$$
$$\frac{dv_y}{dt} = -g - \frac{k}{m} v_y \sqrt{v_x^2 + v_y^2}$$

$$x(0) = x_0, \quad y(0) = y_0, \quad v_x(0) = v_{x0}, \quad v_y(0) = v_{y0}$$

- v_x: The horizontal component of the ball's velocity.
- v_y: The vertical component of the ball's velocity.
- g: The acceleration due to gravity $\approx 9.81\ m/s^2$ on Earth's surface.
- k: The drag coefficient, incorporates air resistance in the model.
- m: The mass of the ball.

Many-body gravitational problem

Problem 1.7

- $M_1, M_2, ...M_n$ mutually attracting masses

Many-body gravitational problem

Problem 1.7

- $M_1, M_2, ...M_n$ mutually attracting masses
- $y_1(t), y_2(t), ..., y_n(t)$ coordinates of masses at time t

Many-body gravitational problem

Problem 1.7

- $M_1, M_2, ...M_n$ mutually attracting masses
- $y_1(t), y_2(t), ..., y_n(t)$ coordinates of masses at time t
- Predict evolution of $y_1(t), y_2(t), ..., y_n(t)$ in time

Many-body gravitational problem

Problem 1.7

- $M_1, M_2, ...M_n$ mutually attracting masses
- $y_1(t), y_2(t), ..., y_n(t)$ coordinates of masses at time t
- Predict evolution of $y_1(t), y_2(t), ..., y_n(t)$ in time

System of seond order diferential equations, J.C.Butcher, Wiley 2016

$$\frac{d^2 y_i(t)}{dt^2} = \sum_{i \neq j} \frac{\gamma M_j (y_i - y_j)}{\|y_i - y_j\|^3}, i = 1, 2, ..., n,$$

$\gamma -$ *gravitational constatnt*

Helmholtz Equation

- The Helmholtz equation arises from the time-independent wave equation.

Helmholtz Equation

- The Helmholtz equation arises from the time-independent wave equation.
- $$\nabla^2 u + k^2 u = 0$$

Helmholtz Equation

- The Helmholtz equation arises from the time-independent wave equation.
- $$\nabla^2 u + k^2 u = 0$$

  where u is the wave amplitude, and k is the wavenumber.

Helmholtz Equation

- The Helmholtz equation arises from the time-independent wave equation.
- $$\nabla^2 u + k^2 u = 0$$

  where u is the wave amplitude, and k is the wavenumber.
- Solutions in specific coordinate systems (e.g., Cartesian, cylindrical, spherical).

Helmholtz Equation

- The Helmholtz equation arises from the time-independent wave equation.
- $$\nabla^2 u + k^2 u = 0$$

  where u is the wave amplitude, and k is the wavenumber.
- Solutions in specific coordinate systems (e.g., Cartesian, cylindrical, spherical).
- In many cases separation of variables reduces the Helmholtz equation to ordinary differential equations (ODEs).

Separation of Variables and ODEs

- Example: Cartesian coordinates in 2D, $u(x, y) = X(x)Y(y)$.

Separation of Variables and ODEs

- Example: Cartesian coordinates in 2D, $u(x, y) = X(x)Y(y)$.
  - Substituting into the Helmholtz equation:

$$\frac{d^2X}{dx^2}Y + X\frac{d^2Y}{dy^2} + k^2XY = 0$$

Separation of Variables and ODEs

- Example: Cartesian coordinates in 2D, $u(x, y) = X(x)Y(y)$.
  - Substituting into the Helmholtz equation:

$$\frac{d^2X}{dx^2}Y + X\frac{d^2Y}{dy^2} + k^2XY = 0$$

  - Dividing by XY and rearranging:

$$\frac{1}{X}\frac{d^2X}{dx^2} + \frac{1}{Y}\frac{d^2Y}{dy^2} + k^2 = 0$$

Separation of Variables and ODEs

- Example: Cartesian coordinates in 2D, $u(x, y) = X(x)Y(y)$.
  - Substituting into the Helmholtz equation:

$$\frac{d^2X}{dx^2}Y + X\frac{d^2Y}{dy^2} + k^2XY = 0$$

  - Dividing by XY and rearranging:

$$\frac{1}{X}\frac{d^2X}{dx^2} + \frac{1}{Y}\frac{d^2Y}{dy^2} + k^2 = 0$$

  - $\Rightarrow$ two ODEs:

$$\frac{d^2X}{dx^2} + k_x^2X = 0$$
$$\frac{d^2Y}{dy^2} + k_y^2Y = 0$$

    where $k_x^2 + k_y^2 = k^2$.

- Similar procedures can be applied in other coordinate systems, resulting in different ODEs (e.g., Bessel's equation in cylindrical coordinates, Legendre's equation in spherical coordinates).

Separation of Variables and ODEs

- Example: Cartesian coordinates in 2D, $u(x, y) = X(x)Y(y)$.
  - Substituting into the Helmholtz equation:

$$\frac{d^2X}{dx^2}Y + X\frac{d^2Y}{dy^2} + k^2XY = 0$$

  - Dividing by XY and rearranging:

$$\frac{1}{X}\frac{d^2X}{dx^2} + \frac{1}{Y}\frac{d^2Y}{dy^2} + k^2 = 0$$

  - $\Rightarrow$ two ODEs:

$$\frac{d^2X}{dx^2} + k_x^2X = 0$$
$$\frac{d^2Y}{dy^2} + k_y^2Y = 0$$

Separation of Variables and ODEs

- Example: Cartesian coordinates in 2D, $u(x, y) = X(x)Y(y)$.
  - Substituting into the Helmholtz equation:

$$\frac{d^2X}{dx^2}Y + X\frac{d^2Y}{dy^2} + k^2XY = 0$$

  - Dividing by XY and rearranging:

$$\frac{1}{X}\frac{d^2X}{dx^2} + \frac{1}{Y}\frac{d^2Y}{dy^2} + k^2 = 0$$

  - $\Rightarrow$ two ODEs:

$$\frac{d^2X}{dx^2} + k_x^2X = 0$$
$$\frac{d^2Y}{dy^2} + k_y^2Y = 0$$

where $k_x^2 + k_y^2 = k^2$.

Separation of Variables and ODEs

- Example: Cartesian coordinates in 2D, $u(x, y) = X(x)Y(y)$.
  - Substituting into the Helmholtz equation:

$$\frac{d^2X}{dx^2}Y + X\frac{d^2Y}{dy^2} + k^2XY = 0$$

  - Dividing by XY and rearranging:

$$\frac{1}{X}\frac{d^2X}{dx^2} + \frac{1}{Y}\frac{d^2Y}{dy^2} + k^2 = 0$$

  - $\Rightarrow$ two ODEs:

$$\frac{d^2X}{dx^2} + k_x^2X = 0$$
$$\frac{d^2Y}{dy^2} + k_y^2Y = 0$$

    where $k_x^2 + k_y^2 = k^2$.

- Similar procedures can be applied in other coordinate systems, resulting in different ODEs (e.g., Bessel's equation in cylindrical coordinates, Legendre's equation in spherical coordinates).

3D Helmholtz Equation and ODEs

- 3D Cartesian coordinates, $u(x, y, z) = X(x)Y(y)Z(z)$.

3D Helmholtz Equation and ODEs

- 3D Cartesian coordinates, $u(x, y, z) = X(x)Y(y)Z(z)$.
- Substituting into the Helmholtz equation:

$$\frac{d^2X}{dx^2}YZ + X\frac{d^2Y}{dy^2}Z + XY\frac{d^2Z}{dz^2} + k^2XYZ = 0$$

3D Helmholtz Equation and ODEs

- 3D Cartesian coordinates, $u(x, y, z) = X(x)Y(y)Z(z)$.
- Substituting into the Helmholtz equation:

$$\frac{d^2X}{dx^2}YZ + X\frac{d^2Y}{dy^2}Z + XY\frac{d^2Z}{dz^2} + k^2XYZ = 0$$

- Dividing by XYZ and rearranging:

$$\frac{1}{X}\frac{d^2X}{dx^2} + \frac{1}{Y}\frac{d^2Y}{dy^2} + \frac{1}{Z}\frac{d^2Z}{dz^2} + k^2 = 0$$

3D Helmholtz Equation and ODEs

- 3D Cartesian coordinates, $u(x, y, z) = X(x)Y(y)Z(z)$.
- Substituting into the Helmholtz equation:

$$\frac{d^2X}{dx^2}YZ + X\frac{d^2Y}{dy^2}Z + XY\frac{d^2Z}{dz^2} + k^2XYZ = 0$$

- Dividing by XYZ and rearranging:

$$\frac{1}{X}\frac{d^2X}{dx^2} + \frac{1}{Y}\frac{d^2Y}{dy^2} + \frac{1}{Z}\frac{d^2Z}{dz^2} + k^2 = 0$$

- $\Rightarrow$ three ODEs:

$$\frac{d^2X}{dx^2} + k_x^2X = 0$$
$$\frac{d^2Y}{dy^2} + k_y^2Y = 0$$
$$\frac{d^2Z}{dz^2} + k_z^2Z = 0$$

  where $k_x^2 + k_y^2 + k_z^2 = k^2$.
- In spherical coordinates, the separation leads to radial, azimuthal, and polar equations. The radial equation is often a spherical Bessel equation, and the angular equations involve associated Legendre polynomials.

3D Helmholtz Equation and ODEs

- 3D Cartesian coordinates, $u(x, y, z) = X(x)Y(y)Z(z)$.
- Substituting into the Helmholtz equation:

$$\frac{d^2X}{dx^2}YZ + X\frac{d^2Y}{dy^2}Z + XY\frac{d^2Z}{dz^2} + k^2XYZ = 0$$

- Dividing by XYZ and rearranging:

$$\frac{1}{X}\frac{d^2X}{dx^2} + \frac{1}{Y}\frac{d^2Y}{dy^2} + \frac{1}{Z}\frac{d^2Z}{dz^2} + k^2 = 0$$

- $\Rightarrow$ three ODEs:

$$\frac{d^2X}{dx^2} + k_x^2X = 0$$

$$\frac{d^2Z}{dz^2} + k_z^2Z = 0$$

where $k_x^2 + k_y^2 + k_z^2 = k^2$.

- In spherical coordinates, the separation leads to radial, azimuthal, and polar equations. The radial equation is often a spherical Bessel equation, and the angular equations involve associated Legendre polynomials.

3D Helmholtz Equation and ODEs

- 3D Cartesian coordinates, $u(x, y, z) = X(x)Y(y)Z(z)$.
- Substituting into the Helmholtz equation:

$$\frac{d^2X}{dx^2}YZ + X\frac{d^2Y}{dy^2}Z + XY\frac{d^2Z}{dz^2} + k^2XYZ = 0$$

- Dividing by XYZ and rearranging:

$$\frac{1}{X}\frac{d^2X}{dx^2} + \frac{1}{Y}\frac{d^2Y}{dy^2} + \frac{1}{Z}\frac{d^2Z}{dz^2} + k^2 = 0$$

- $\Rightarrow$ three ODEs:

$$\frac{d^2X}{dx^2} + k_x^2X = 0$$
$$\frac{d^2Y}{dy^2} + k_y^2Y = 0$$
$$\frac{d^2Z}{dz^2} + k_z^2Z = 0$$

3D Helmholtz Equation and ODEs

- 3D Cartesian coordinates, $u(x, y, z) = X(x)Y(y)Z(z)$.
- Substituting into the Helmholtz equation:

$$\frac{d^2X}{dx^2}YZ + X\frac{d^2Y}{dy^2}Z + XY\frac{d^2Z}{dz^2} + k^2XYZ = 0$$

- Dividing by XYZ and rearranging:

$$\frac{1}{X}\frac{d^2X}{dx^2} + \frac{1}{Y}\frac{d^2Y}{dy^2} + \frac{1}{Z}\frac{d^2Z}{dz^2} + k^2 = 0$$

- $\Rightarrow$ three ODEs:

$$\frac{d^2X}{dx^2} + k_x^2X = 0$$
$$\frac{d^2Y}{dy^2} + k_y^2Y = 0$$
$$\frac{d^2Z}{dz^2} + k_z^2Z = 0$$

where $k_x^2 + k_y^2 + k_z^2 = k^2$.

3D Helmholtz Equation and ODEs

- 3D Cartesian coordinates, $u(x, y, z) = X(x)Y(y)Z(z)$.
- Substituting into the Helmholtz equation:

$$\frac{d^2X}{dx^2}YZ + X\frac{d^2Y}{dy^2}Z + XY\frac{d^2Z}{dz^2} + k^2XYZ = 0$$

- Dividing by XYZ and rearranging:

$$\frac{1}{X}\frac{d^2X}{dx^2} + \frac{1}{Y}\frac{d^2Y}{dy^2} + \frac{1}{Z}\frac{d^2Z}{dz^2} + k^2 = 0$$

- $\Rightarrow$ three ODEs:

$$\frac{d^2X}{dx^2} + k_x^2X = 0$$
$$\frac{d^2Y}{dy^2} + k_y^2Y = 0$$
$$\frac{d^2Z}{dz^2} + k_z^2Z = 0$$

  where $k_x^2 + k_y^2 + k_z^2 = k^2$.
- In spherical coordinates, the separation leads to radial, azimuthal, and polar equations. The radial equation is often a spherical Bessel equation, and the angular equations involve associated Legendre polynomials.

Schrödinger Equation

- The time-independent Schrödinger equation describes the stationary states of a quantum mechanical system.

Schrödinger Equation

- The time-independent Schrödinger equation describes the stationary states of a quantum mechanical system.
- It is given by:

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V(\mathbf{r})\psi = E\psi$$

Schrödinger Equation

- The time-independent Schrödinger equation describes the stationary states of a quantum mechanical system.
- It is given by:

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V(\mathbf{r})\psi = E\psi$$

where ψ is the wavefunction, $V(\mathbf{r})$ is the potential, E is the energy, $\hbar$ is the reduced Planck constant, and m is the mass.

Schrödinger Equation

- The time-independent Schrödinger equation describes the stationary states of a quantum mechanical system.
- It is given by:

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V(\mathbf{r})\psi = E\psi$$

  where ψ is the wavefunction, $V(\mathbf{r})$ is the potential, E is the energy, $\hbar$ is the reduced Planck constant, and m is the mass.
- Solutions in specific coordinate systems based on the symmetry of the potential $V(\mathbf{r})$.

Separation of Variables and ODEs (Hydrogen Atom)

- For the hydrogen atom, $V(r) = -\frac{e^2}{4\pi\epsilon_0 r}$, which is spherically symmetric.

Separation of Variables and ODEs (Hydrogen Atom)

- For the hydrogen atom, $V(r) = -\frac{e^2}{4\pi\epsilon_0 r}$, which is spherically symmetric.
- Spherical coordinates (r, θ, ϕ) and a separable solution:

$$\psi(r, \theta, \phi) = R(r)Y(\theta, \phi)$$

Separation of Variables and ODEs (Hydrogen Atom)

- For the hydrogen atom, $V(r) = -\frac{e^2}{4\pi\epsilon_0 r}$, which is spherically symmetric.
- Spherical coordinates (r, θ, ϕ) and a separable solution:

$$\psi(r, \theta, \phi) = R(r)Y(\theta, \phi)$$

- Substituting into the Schrödinger equation:
  - Radial equation:

$$-\frac{\hbar^2}{2m}\left(\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)\right) + \left(V(r) + \frac{\hbar^2 l(l+1)}{2mr^2}\right)R = ER$$

Separation of Variables and ODEs (Hydrogen Atom)

- For the hydrogen atom, $V(r) = -\frac{e^2}{4\pi\epsilon_0 r}$, which is spherically symmetric.
- Spherical coordinates (r, θ, ϕ) and a separable solution:

$$\psi(r, \theta, \phi) = R(r)Y(\theta, \phi)$$

- Substituting into the Schrödinger equation:
  - Radial equation:

$$-\frac{\hbar^2}{2m}\left(\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)\right) + \left(V(r) + \frac{\hbar^2 l(l+1)}{2mr^2}\right)R = ER$$

  - Angular equation (spherical harmonics):

$$\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial Y}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2 Y}{\partial\phi^2} + l(l+1)Y = 0$$

Separation of Variables and ODEs (Hydrogen Atom)

- For the hydrogen atom, $V(r) = -\frac{e^2}{4\pi\epsilon_0 r}$, which is spherically symmetric.
- Spherical coordinates (r, θ, ϕ) and a separable solution:

$$\psi(r, \theta, \phi) = R(r)Y(\theta, \phi)$$

- Substituting into the Schrödinger equation:
  - Radial equation:

$$-\frac{\hbar^2}{2m}\left(\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)\right) + \left(V(r) + \frac{\hbar^2 l(l+1)}{2mr^2}\right)R = ER$$

  - Angular equation (spherical harmonics):

$$\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial Y}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2 Y}{\partial\phi^2} + l(l+1)Y = 0$$

- The radial equation is an ODE for $R(r)$, and the angular equation is solved by spherical harmonics $Y(\theta, \phi)$.

Other Potentials and ODEs

- Harmonic oscillator: $V(x) = \frac{1}{2} m\omega^2 x^2$ in 1D.

Other Potentials and ODEs

- Harmonic oscillator: $V(x) = \frac{1}{2}m\omega^2 x^2$ in 1D.
- The Schrödinger equation becomes:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + \frac{1}{2}m\omega^2 x^2\psi = E\psi$$

Other Potentials and ODEs

- Harmonic oscillator: $V(x) = \frac{1}{2}m\omega^2 x^2$ in 1D.
- The Schrödinger equation becomes:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + \frac{1}{2}m\omega^2 x^2\psi = E\psi$$

- This leads to Hermite's differential equation, and the solutions are Hermite polynomials multiplied by a Gaussian.

Other Potentials and ODEs

- Harmonic oscillator: $V(x) = \frac{1}{2}m\omega^2 x^2$ in 1D.
- The Schrödinger equation becomes:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + \frac{1}{2}m\omega^2 x^2\psi = E\psi$$

- This leads to Hermite's differential equation, and the solutions are Hermite polynomials multiplied by a Gaussian.
- Particle in a box: $V(x) = 0$ inside the box, $V(x) = \infty$ outside.

Other Potentials and ODEs

- Harmonic oscillator: $V(x) = \frac{1}{2}m\omega^2 x^2$ in 1D.
- The Schrödinger equation becomes:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + \frac{1}{2}m\omega^2 x^2\psi = E\psi$$

- This leads to Hermite's differential equation, and the solutions are Hermite polynomials multiplied by a Gaussian.
- Particle in a box: $V(x) = 0$ inside the box, $V(x) = \infty$ outside.
- The Schrödinger equation inside the box is:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi$$

Other Potentials and ODEs

- Harmonic oscillator: $V(x) = \frac{1}{2}m\omega^2 x^2$ in 1D.
- The Schrödinger equation becomes:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + \frac{1}{2}m\omega^2 x^2\psi = E\psi$$

- This leads to Hermite's differential equation, and the solutions are Hermite polynomials multiplied by a Gaussian.
- Particle in a box: $V(x) = 0$ inside the box, $V(x) = \infty$ outside.
- The Schrödinger equation inside the box is:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi$$

- This is a simple second-order ODE with sinusoidal solutions, subject to boundary conditions.

Other Potentials and ODEs

- Harmonic oscillator: $V(x) = \frac{1}{2}m\omega^2 x^2$ in 1D.
- The Schrödinger equation becomes:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + \frac{1}{2}m\omega^2 x^2\psi = E\psi$$

- This leads to Hermite's differential equation, and the solutions are Hermite polynomials multiplied by a Gaussian.
- Particle in a box: $V(x) = 0$ inside the box, $V(x) = \infty$ outside.
- The Schrödinger equation inside the box is:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi$$

- This is a simple second-order ODE with sinusoidal solutions, subject to boundary conditions.
- The choice of coordinate system and separation of variables depends heavily on the form of the potential $V(\mathbf{r})$.

More Potential Functions

- Coulomb potential in 1D: $V(x) = -\frac{e^2}{4\pi\epsilon_0|x|}$.

More Potential Functions

- Coulomb potential in 1D: $V(x) = -\frac{e^2}{4\pi\epsilon_0|x|}$.
- This is a simplified model of the hydrogen atom in one dimension.

More Potential Functions

- Coulomb potential in 1D: $V(x) = -\frac{e^2}{4\pi\epsilon_0|x|}$.
- This is a simplified model of the hydrogen atom in one dimension.
- The solutions involve confluent hypergeometric functions.

More Potential Functions

- Coulomb potential in 1D: $V(x) = -\frac{e^2}{4\pi\epsilon_0|x|}$.
- This is a simplified model of the hydrogen atom in one dimension.
- The solutions involve confluent hypergeometric functions.
- Finite square well: $V(x) = -V_0$ for $|x| < a$, $V(x) = 0$ for $|x| > a$.

More Potential Functions

- Coulomb potential in 1D: $V(x) = -\frac{e^2}{4\pi\epsilon_0|x|}$.
- This is a simplified model of the hydrogen atom in one dimension.
- The solutions involve confluent hypergeometric functions.
- Finite square well: $V(x) = -V_0$ for $|x| < a$, $V(x) = 0$ for $|x| > a$.
- This models a potential with finite depth, leading to bound and scattering states.

More Potential Functions

- Coulomb potential in 1D: $V(x) = -\frac{e^2}{4\pi\epsilon_0|x|}$.
- This is a simplified model of the hydrogen atom in one dimension.
- The solutions involve confluent hypergeometric functions.
- Finite square well: $V(x) = -V_0$ for $|x| < a$, $V(x) = 0$ for $|x| > a$.
- This models a potential with finite depth, leading to bound and scattering states.
- The solutions are piecewise defined, with sinusoidal and exponential forms matched at the boundaries.

More Potential Functions

- Coulomb potential in 1D: $V(x) = -\frac{e^2}{4\pi\epsilon_0|x|}$.
- This is a simplified model of the hydrogen atom in one dimension.
- The solutions involve confluent hypergeometric functions.
- Finite square well: $V(x) = -V_0$ for $|x| < a$, $V(x) = 0$ for $|x| > a$.
- This models a potential with finite depth, leading to bound and scattering states.
- The solutions are piecewise defined, with sinusoidal and exponential forms matched at the boundaries.
- Morse potential: $V(r) = D_e(1 - e^{-a(r-r_e)})^2$.

More Potential Functions

- Coulomb potential in 1D: $V(x) = -\frac{e^2}{4\pi\epsilon_0|x|}$.
- This is a simplified model of the hydrogen atom in one dimension.
- The solutions involve confluent hypergeometric functions.
- Finite square well: $V(x) = -V_0$ for $|x| < a$, $V(x) = 0$ for $|x| > a$.
- This models a potential with finite depth, leading to bound and scattering states.
- The solutions are piecewise defined, with sinusoidal and exponential forms matched at the boundaries.
- Morse potential: $V(r) = D_e(1 - e^{-a(r-r_e)})^2$.
- This models the vibrational potential of diatomic molecules.

More Potential Functions

- Coulomb potential in 1D: $V(x) = -\frac{e^2}{4\pi\epsilon_0|x|}$.
- This is a simplified model of the hydrogen atom in one dimension.
- The solutions involve confluent hypergeometric functions.
- Finite square well: $V(x) = -V_0$ for $|x| < a$, $V(x) = 0$ for $|x| > a$.
- This models a potential with finite depth, leading to bound and scattering states.
- The solutions are piecewise defined, with sinusoidal and exponential forms matched at the boundaries.
- Morse potential: $V(r) = D_e(1 - e^{-a(r-r_e)})^2$.
- This models the vibrational potential of diatomic molecules.
- The solutions involve Laguerre polynomials.

More Potential Functions

- Coulomb potential in 1D: $V(x) = -\frac{e^2}{4\pi\epsilon_0|x|}$.
- This is a simplified model of the hydrogen atom in one dimension.
- The solutions involve confluent hypergeometric functions.
- Finite square well: $V(x) = -V_0$ for $|x| < a$, $V(x) = 0$ for $|x| > a$.
- This models a potential with finite depth, leading to bound and scattering states.
- The solutions are piecewise defined, with sinusoidal and exponential forms matched at the boundaries.
- Morse potential: $V(r) = D_e(1 - e^{-a(r-r_e)})^2$.
- This models the vibrational potential of diatomic molecules.
- The solutions involve Laguerre polynomials.
- Delta function potential: $V(x) = \lambda\delta(x)$.

More Potential Functions

- Coulomb potential in 1D: $V(x) = -\frac{e^2}{4\pi\epsilon_0|x|}$.
- This is a simplified model of the hydrogen atom in one dimension.
- The solutions involve confluent hypergeometric functions.
- Finite square well: $V(x) = -V_0$ for $|x| < a$, $V(x) = 0$ for $|x| > a$.
- This models a potential with finite depth, leading to bound and scattering states.
- The solutions are piecewise defined, with sinusoidal and exponential forms matched at the boundaries.
- Morse potential: $V(r) = D_e(1 - e^{-a(r-r_e)})^2$.
- This models the vibrational potential of diatomic molecules.
- The solutions involve Laguerre polynomials.
- Delta function potential: $V(x) = \lambda\delta(x)$.
- This is a simplified model of a very short-range potential, leading to a bound state

More Potential Functions

- Coulomb potential in 1D: $V(x) = -\frac{e^2}{4\pi\epsilon_0|x|}$.
- This is a simplified model of the hydrogen atom in one dimension.
- The solutions involve confluent hypergeometric functions.
- Finite square well: $V(x) = -V_0$ for $|x| < a$, $V(x) = 0$ for $|x| > a$.
- This models a potential with finite depth, leading to bound and scattering states.
- The solutions are piecewise defined, with sinusoidal and exponential forms matched at the boundaries.
- Morse potential: $V(r) = D_e(1 - e^{-a(r-r_e)})^2$.
- This models the vibrational potential of diatomic molecules.
- The solutions involve Laguerre polynomials.
- Delta function potential: $V(x) = \lambda\delta(x)$.
- This is a simplified model of a very short-range potential, leading to a bound state
- The solution involves a discontinuity in the derivative of the wavefunction at $x = 0$.

Boundary Conditions and Infinite Domains

- Boundary conditions are crucial for determining physically acceptable solutions to the Schrödinger equation.

Boundary Conditions and Infinite Domains

- Boundary conditions are crucial for determining physically acceptable solutions to the Schrödinger equation.
- For bound states, the wavefunction must vanish at infinity: $\lim_{|x|\to\infty} \psi(x) = 0$.

Boundary Conditions and Infinite Domains

- Boundary conditions are crucial for determining physically acceptable solutions to the Schrödinger equation.
- For bound states, the wavefunction must vanish at infinity: $\lim_{|x|\to\infty} \psi(x) = 0$.
- Continuity of the wavefunction and its derivative (where the potential is finite) is required.

Boundary Conditions and Infinite Domains

- Boundary conditions are crucial for determining physically acceptable solutions to the Schrödinger equation.
- For bound states, the wavefunction must vanish at infinity: $\lim_{|x|\to\infty} \psi(x) = 0$.
- Continuity of the wavefunction and its derivative (where the potential is finite) is required.
- In finite domains (like the particle in a box), boundary conditions are imposed at the edges of the domain.

Boundary Conditions and Infinite Domains

- Boundary conditions are crucial for determining physically acceptable solutions to the Schrödinger equation.
- For bound states, the wavefunction must vanish at infinity: $\lim_{|x|\to\infty} \psi(x) = 0$.
- Continuity of the wavefunction and its derivative (where the potential is finite) is required.
- In finite domains (like the particle in a box), boundary conditions are imposed at the edges of the domain.
- For infinite domains, the behavior of the wavefunction at infinity dictates the allowed energy levels and wavefunctions.

Boundary Conditions and Infinite Domains

- Boundary conditions are crucial for determining physically acceptable solutions to the Schrödinger equation.
- For bound states, the wavefunction must vanish at infinity: $\lim_{|x|\to\infty} \psi(x) = 0$.
- Continuity of the wavefunction and its derivative (where the potential is finite) is required.
- In finite domains (like the particle in a box), boundary conditions are imposed at the edges of the domain.
- For infinite domains, the behavior of the wavefunction at infinity dictates the allowed energy levels and wavefunctions.
- The requirement of normalizability, $\int |\psi|^2 d\mathbf{r} < \infty$, is essential for physical solutions.

Boundary Conditions and Infinite Domains

- Boundary conditions are crucial for determining physically acceptable solutions to the Schrödinger equation.
- For bound states, the wavefunction must vanish at infinity: $\lim_{|x|\to\infty} \psi(x) = 0$.
- Continuity of the wavefunction and its derivative (where the potential is finite) is required.
- In finite domains (like the particle in a box), boundary conditions are imposed at the edges of the domain.
- For infinite domains, the behavior of the wavefunction at infinity dictates the allowed energy levels and wavefunctions.
- The requirement of normalizability, $\int |\psi|^2 d\mathbf{r} < \infty$, is essential for physical solutions.
- Scattering states, where particles come from and go to infinity, have different boundary conditions than bound states.

Domain Transformations

- Sometimes, it's useful to transform an infinite domain to a finite one for numerical or analytical convenience.

Domain Transformations

- Sometimes, it's useful to transform an infinite domain to a finite one for numerical or analytical convenience.
- For example, the infinite domain $x \in (-\infty, \infty)$ can be mapped to $y \in (-1, 1)$ using $x = \tan(\frac{\pi y}{2})$.

Domain Transformations

- Sometimes, it's useful to transform an infinite domain to a finite one for numerical or analytical convenience.
- For example, the infinite domain $x \in (-\infty, \infty)$ can be mapped to $y \in (-1, 1)$ using $x = \tan(\frac{\pi y}{2})$.
- This transformation changes the derivatives: $\frac{d}{dx} = \frac{\pi}{2} \sec^2(\frac{\pi y}{2}) \frac{d}{dy}$.

Domain Transformations

- Sometimes, it's useful to transform an infinite domain to a finite one for numerical or analytical convenience.
- For example, the infinite domain $x \in (-\infty, \infty)$ can be mapped to $y \in (-1, 1)$ using $x = \tan(\frac{\pi y}{2})$.
- This transformation changes the derivatives: $\frac{d}{dx} = \frac{\pi}{2} \sec^2(\frac{\pi y}{2}) \frac{d}{dy}$.
- Substituting this into the Schrödinger equation will result in a new ODE with coefficients that depend on y.

Domain Transformations

- Sometimes, it's useful to transform an infinite domain to a finite one for numerical or analytical convenience.
- For example, the infinite domain $x \in (-\infty, \infty)$ can be mapped to $y \in (-1, 1)$ using $x = \tan(\frac{\pi y}{2})$.
- This transformation changes the derivatives: $\frac{d}{dx} = \frac{\pi}{2} \sec^2(\frac{\pi y}{2}) \frac{d}{dy}$.
- Substituting this into the Schrödinger equation will result in a new ODE with coefficients that depend on y.
- This often leads to more complex coefficients, but it allows the use of numerical methods suitable for finite domains.

Domain Transformations

- Sometimes, it's useful to transform an infinite domain to a finite one for numerical or analytical convenience.
- For example, the infinite domain $x \in (-\infty, \infty)$ can be mapped to $y \in (-1, 1)$ using $x = \tan(\frac{\pi y}{2})$.
- This transformation changes the derivatives: $\frac{d}{dx} = \frac{\pi}{2} \sec^2(\frac{\pi y}{2}) \frac{d}{dy}$.
- Substituting this into the Schrödinger equation will result in a new ODE with coefficients that depend on y.
- This often leads to more complex coefficients, but it allows the use of numerical methods suitable for finite domains.
- Another transformation, for the radial coordinate $r \in [0, \infty)$, is $r = \frac{Lz}{1-z}$, mapping $z \in [0, 1)$.

Domain Transformations

- Sometimes, it's useful to transform an infinite domain to a finite one for numerical or analytical convenience.
- For example, the infinite domain $x \in (-\infty, \infty)$ can be mapped to $y \in (-1, 1)$ using $x = \tan(\frac{\pi y}{2})$.
- This transformation changes the derivatives: $\frac{d}{dx} = \frac{\pi}{2} \sec^2(\frac{\pi y}{2}) \frac{d}{dy}$.
- Substituting this into the Schrödinger equation will result in a new ODE with coefficients that depend on y.
- This often leads to more complex coefficients, but it allows the use of numerical methods suitable for finite domains.
- Another transformation, for the radial coordinate $r \in [0, \infty)$, is $r = \frac{Lz}{1-z}$, mapping $z \in [0, 1)$.
- The transformed ODE will have new coefficients dependent on z, and the boundary conditions change accordingly.

Schrödinger Equation: Finite Domain Transformation

- Consider the 1D Schrödinger equation:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi = E\psi$$

Schrödinger Equation: Finite Domain Transformation

- Consider the 1D Schrödinger equation:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi = E\psi$$

- Assume $x \in (-\infty, \infty)$, an infinite domain.

Schrödinger Equation: Finite Domain Transformation

- Consider the 1D Schrödinger equation:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi = E\psi$$

- Assume $x \in (-\infty, \infty)$, an infinite domain.
- We want to transform it to a finite domain, say $y \in (-1, 1)$.

Schrödinger Equation: Finite Domain Transformation

- Consider the 1D Schrödinger equation:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi = E\psi$$

- Assume $x \in (-\infty, \infty)$, an infinite domain.
- We want to transform it to a finite domain, say $y \in (-1, 1)$.
- Let's use the transformation: $x = \tan\left(\frac{\pi y}{2}\right)$.

Schrödinger Equation: Finite Domain Transformation

- Consider the 1D Schrödinger equation:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi = E\psi$$

- Assume $x \in (-\infty, \infty)$, an infinite domain.
- We want to transform it to a finite domain, say $y \in (-1, 1)$.
- Let's use the transformation: $x = \tan\left(\frac{\pi y}{2}\right)$.
- The derivative transformation is:

$$\frac{d}{dx} = \frac{\pi}{2}\sec^2\left(\frac{\pi y}{2}\right)\frac{d}{dy}$$

Schrödinger Equation: Finite Domain Transformation

- Consider the 1D Schrödinger equation:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi = E\psi$$

- Assume $x \in (-\infty, \infty)$, an infinite domain.
- We want to transform it to a finite domain, say $y \in (-1, 1)$.
- Let's use the transformation: $x = \tan\left(\frac{\pi y}{2}\right)$.
- The derivative transformation is:

$$\frac{d}{dx} = \frac{\pi}{2}\sec^2\left(\frac{\pi y}{2}\right)\frac{d}{dy}$$

- Applying this twice for the second derivative:

$$\frac{d^2}{dx^2} = \left(\frac{\pi}{2}\sec^2\left(\frac{\pi y}{2}\right)\right)^2\frac{d^2}{dy^2} + \left(\frac{\pi^2}{2}\sec^2\left(\frac{\pi y}{2}\right)\tan\left(\frac{\pi y}{2}\right)\right)\frac{d}{dy}$$

Schrödinger Equation: Finite Domain Transformation

- Consider the 1D Schrödinger equation:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi = E\psi$$

- Assume $x \in (-\infty, \infty)$, an infinite domain.
- We want to transform it to a finite domain, say $y \in (-1, 1)$.
- Let's use the transformation: $x = \tan\left(\frac{\pi y}{2}\right)$.
- The derivative transformation is:

$$\frac{d}{dx} = \frac{\pi}{2}\sec^2\left(\frac{\pi y}{2}\right)\frac{d}{dy}$$

- Applying this twice for the second derivative:

$$\frac{d^2}{dx^2} = \left(\frac{\pi}{2}\sec^2\left(\frac{\pi y}{2}\right)\right)^2\frac{d^2}{dy^2} + \left(\frac{\pi^2}{2}\sec^2\left(\frac{\pi y}{2}\right)\tan\left(\frac{\pi y}{2}\right)\right)\frac{d}{dy}$$

- Substituting into the Schrödinger equation, we obtain a new ODE for $\psi(y)$:

$$-\frac{\hbar^2}{2m}\left[\left(\frac{\pi}{2}\sec^2\left(\frac{\pi y}{2}\right)\right)^2\frac{d^2\psi}{dy^2} + \left(\frac{\pi^2}{2}\sec^2\left(\frac{\pi y}{2}\right)\tan\left(\frac{\pi y}{2}\right)\right)\frac{d\psi}{dy}\right] + V\left(\tan\left(\frac{\pi y}{2}\right)\right)\psi = E\psi$$

Schrödinger Equation: Finite Domain Transformation

- Consider the 1D Schrödinger equation:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi = E\psi$$

- Assume $x \in (-\infty, \infty)$, an infinite domain.
- We want to transform it to a finite domain, say $y \in (-1, 1)$.
- Let's use the transformation: $x = \tan\left(\frac{\pi y}{2}\right)$.
- The derivative transformation is:

$$\frac{d}{dx} = \frac{\pi}{2}\sec^2\left(\frac{\pi y}{2}\right)\frac{d}{dy}$$

- Applying this twice for the second derivative:

$$\frac{d^2}{dx^2} = \left(\frac{\pi}{2}\sec^2\left(\frac{\pi y}{2}\right)\right)^2\frac{d^2}{dy^2} + \left(\frac{\pi^2}{2}\sec^2\left(\frac{\pi y}{2}\right)\tan\left(\frac{\pi y}{2}\right)\right)\frac{d}{dy}$$

- Substituting into the Schrödinger equation, we obtain a new ODE for $\psi(y)$:

$$-\frac{\hbar^2}{2m}\left[\left(\frac{\pi}{2}\sec^2\left(\frac{\pi y}{2}\right)\right)^2\frac{d^2\psi}{dy^2} + \left(\frac{\pi^2}{2}\sec^2\left(\frac{\pi y}{2}\right)\tan\left(\frac{\pi y}{2}\right)\right)\frac{d\psi}{dy}\right] + V\left(\tan\left(\frac{\pi y}{2}\right)\right)\psi = E\psi$$

- The resulting ODE has variable coefficients depending on y, but the domain is now finite.

Swinging Fabric: Conceptual Model

- Modelling of a piece of fabric hanging from a single point, swinging in a 2D plane.

Swinging Fabric: Conceptual Model

- Modelling of a piece of fabric hanging from a single point, swinging in a 2D plane.
- $\Rightarrow$ Simplification: the complex deformation of cloth viewed as a pendulum-like motion.

Swinging Fabric: Conceptual Model

- Modelling of a piece of fabric hanging from a single point, swinging in a 2D plane.
- $\Rightarrow$ Simplification: the complex deformation of cloth viewed as a pendulum-like motion.
- Key forces involved:
  - Gravity (restoring force).

Swinging Fabric: Conceptual Model

- Modelling of a piece of fabric hanging from a single point, swinging in a 2D plane.
- $\Rightarrow$ Simplification: the complex deformation of cloth viewed as a pendulum-like motion.
- Key forces involved:
  - Gravity (restoring force).
  - Internal damping (friction within the fabric).

Swinging Fabric: Conceptual Model

- Modelling of a piece of fabric hanging from a single point, swinging in a 2D plane.
- $\Rightarrow$ Simplification: the complex deformation of cloth viewed as a pendulum-like motion.
- Key forces involved:
  - Gravity (restoring force).
  - Internal damping (friction within the fabric).
  - Air resistance (drag).

Swinging Fabric: Conceptual Model

- Modelling of a piece of fabric hanging from a single point, swinging in a 2D plane.
- $\Rightarrow$ Simplification: the complex deformation of cloth viewed as a pendulum-like motion.
- Key forces involved:
  - Gravity (restoring force).
  - Internal damping (friction within the fabric).
  - Air resistance (drag).
- We aim to describe the angular motion of the fabric.

Mathematical Model: ODE Derivation

- $\theta(t)$ - the angle of the fabric with respect to the vertical.

Mathematical Model: ODE Derivation

- $\theta(t)$ - the angle of the fabric with respect to the vertical.
- Newton's second law for rotational motion:

$$I\ddot{\theta} = \sum \tau$$

Mathematical Model: ODE Derivation

- $\theta(t)$ - the angle of the fabric with respect to the vertical.
- Newton's second law for rotational motion:

$$I\ddot{\theta} = \sum \tau$$

where I is the moment of inertia and τ are the torques.

Mathematical Model: ODE Derivation

- $\theta(t)$ - the angle of the fabric with respect to the vertical.
- Newton's second law for rotational motion:

$$I\ddot{\theta} = \sum \tau$$

  where I is the moment of inertia and τ are the torques.
- Torques due to:
  - Gravity: $\tau_g = -mgL\sin(\theta)$ (restoring).

Mathematical Model: ODE Derivation

- $\theta(t)$ - the angle of the fabric with respect to the vertical.
- Newton's second law for rotational motion:

$$I\ddot{\theta} = \sum \tau$$

  where I is the moment of inertia and τ are the torques.
- Torques due to:
  - Gravity: $\tau_g = -mgL\sin(\theta)$ (restoring).
  - Damping: $\tau_d = -c\dot{\theta}$.

Mathematical Model: ODE Derivation

- $\theta(t)$ - the angle of the fabric with respect to the vertical.
- Newton's second law for rotational motion:

$$I\ddot{\theta} = \sum \tau$$

  where I is the moment of inertia and τ are the torques.
- Torques due to:
  - Gravity: $\tau_g = -mgL\sin(\theta)$ (restoring).
  - Damping: $\tau_d = -c\dot{\theta}$.
  - Air drag: $\tau_{drag} = -b\dot{\theta}^2$.

Mathematical Model: ODE Derivation

- $\theta(t)$ - the angle of the fabric with respect to the vertical.
- Newton's second law for rotational motion:

$$I\ddot{\theta} = \sum \tau$$

  where I is the moment of inertia and τ are the torques.
- Torques due to:
  - Gravity: $\tau_g = -mgL\sin(\theta)$ (restoring).
  - Damping: $\tau_d = -c\dot{\theta}$.
  - Air drag: $\tau_{drag} = -b\dot{\theta}^2$.

Mathematical Model: ODE Derivation

- $\theta(t)$ - the angle of the fabric with respect to the vertical.
- Newton's second law for rotational motion:

$$I\ddot{\theta} = \sum \tau$$

  where I is the moment of inertia and τ are the torques.
- Torques due to:
  - Gravity: $\tau_g = -mgL\sin(\theta)$ (restoring).
  - Damping: $\tau_d = -c\dot{\theta}$.
  - Air drag: $\tau_{drag} = -b\dot{\theta}^2$.
- $\Rightarrow$ ODE:

$$I\ddot{\theta} + c\dot{\theta} + mgL\sin(\theta) + b\dot{\theta}^2 = 0$$

Mathematical Model: ODE Derivation

- $\theta(t)$ - the angle of the fabric with respect to the vertical.
- Newton's second law for rotational motion:

$$I\ddot{\theta} = \sum \tau$$

  where I is the moment of inertia and τ are the torques.
- Torques due to:
  - Gravity: $\tau_g = -mgL\sin(\theta)$ (restoring).
  - Damping: $\tau_d = -c\dot{\theta}$.
  - Air drag: $\tau_{drag} = -b\dot{\theta}^2$.
- $\Rightarrow$ ODE:

$$I\ddot{\theta} + c\dot{\theta} + mgL\sin(\theta) + b\dot{\theta}^2 = 0$$

- $=$ Nonlinear, second-order ODE describing the fabric's angular motion.

Q & A